

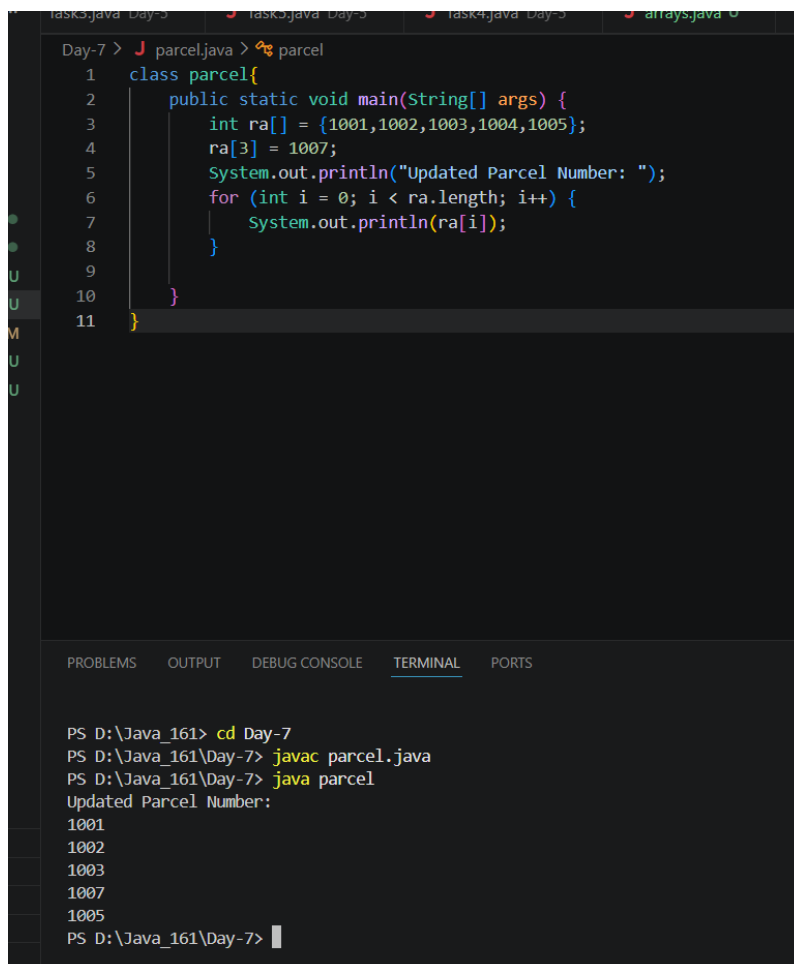
Task 1: Delivery Parcel Tracking System

Scenario:

A delivery shop stores parcel tracking numbers. One parcel number is corrected after update. Then all parcels are displayed.

Requirement:

- Create array
- Add values
- Modify one value
- Display using loop



```
Day-7 > parcel.java > parcel
1  class parcel{
2      public static void main(String[] args) {
3          int ra[] = {1001,1002,1003,1004,1005};
4          ra[3] = 1007;
5          System.out.println("Updated Parcel Number: ");
6          for (int i = 0; i < ra.length; i++) {
7              System.out.println(ra[i]);
8          }
9      }
10 }
11 }
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS D:\Java_161> cd Day-7
PS D:\Java_161\Day-7> javac parcel.java
PS D:\Java_161\Day-7> java parcel
Updated Parcel Number:
1001
1002
1003
1007
1005
PS D:\Java_161\Day-7> 
```

Task 2 : Hospital Patient ID Search System

Scenario:

A hospital stores patient IDs. The system checks whether a patient ID is available for treatment.

Question:

- Store patient IDs in an array
- Search a given patient ID
- Print whether it is found or not

```
Day-7 > J pat.java > pat
1  import java.util.Scanner;
2  class pat{
    Run main | Debug main
3  public static void main(String[] args) {
4      int ra[] = {101,102,103,104,105};
5      System.out.println("Enter the patient ID: ");
6      Scanner sc = new Scanner(System.in);
7      int ra1 = sc.nextInt();
8      int c = 0;
9      for (int i = 0; i < ra.length; i++) {
10         if(ra1==ra[i]){
11             System.out.println("Patient found");
12             c++;
13         }
14     }
15     if(c==0){
16         System.out.println("Patient Not Found");
17     }
18 }
19 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Updated Parcel Number:
1001
1002
1003
1007
1005
PS D:\Java_161\Day-7> javac pat.java
PS D:\Java_161\Day-7> java pat
Enter the patient ID:
101
Patient found
PS D:\Java_161\Day-7> |
```

Task 3: Library Book ID Sorting System

Scenario:

A library stores book IDs in random order. The system needs to arrange them in ascending order.

- Store book IDs in an array
- Sort the IDs in ascending order
- Display sorted list

```
Day-7 > J lib.java > lib
1  import java.util.Arrays;
2  class lib{
    Run main | Debug main
3      public static void main(String[] args) {
4          int ra[] = {401,402,407,408,405,406};
5          Arrays.sort(ra);
6          System.out.println("Book ID's Sorted:");
7          for (int i = 0; i < ra.length; i++) {
8              System.out.println(ra[i]);
9          }
10     }
11 }
```

101
Patient found
PS D:\Java_161\Day-7> javac lib.java
PS D:\Java_161\Day-7> java lib
Book ID's Sorted:
401
402
405
406
407
408
PS D:\Java_161\Day-7> |

Task 4: Number List Copy Using Methods

Scenario:

You have a list of numbers. You want to copy the list using different built-in methods.

- Store numbers in an array
- Copy the array using:
 1. `clone()`
 2. `Arrays.copyOf()`
 3. `System.arraycopy()`

```
Day-7 > J array.java > array
1  import java.util.Arrays;
2  class array{
    Run main | Debug main
3      public static void main(String[] args) {
4          int original[] = {1,2,3,4,5};
5          int clo[] = original.clone();
6          int new1[] = new int[original.length];
7          System.out.println("Cloned Array: ");
8          System.out.println(Arrays.toString(clo));
9          System.out.println("array copy method: ");
10         System.arraycopy(original, 0, new1, 0, 5);
11         System.out.println(Arrays.toString(new1));
12         System.out.println("array copyOf method: ");
13         int arr2[] = Arrays.copyOf(original, 5);
14         System.out.println(Arrays.toString(arr2));
15     }
16 }
```

406
407
408
PS D:\Java_161\Day-7> javac array.java
PS D:\Java_161\Day-7> java array
Cloned Array:
[1, 2, 3, 4, 5]
array copy method:
[1, 2, 3, 4, 5]
array copyOf method:
[1, 2, 3, 4, 5]
PS D:\Java_161\Day-7> |

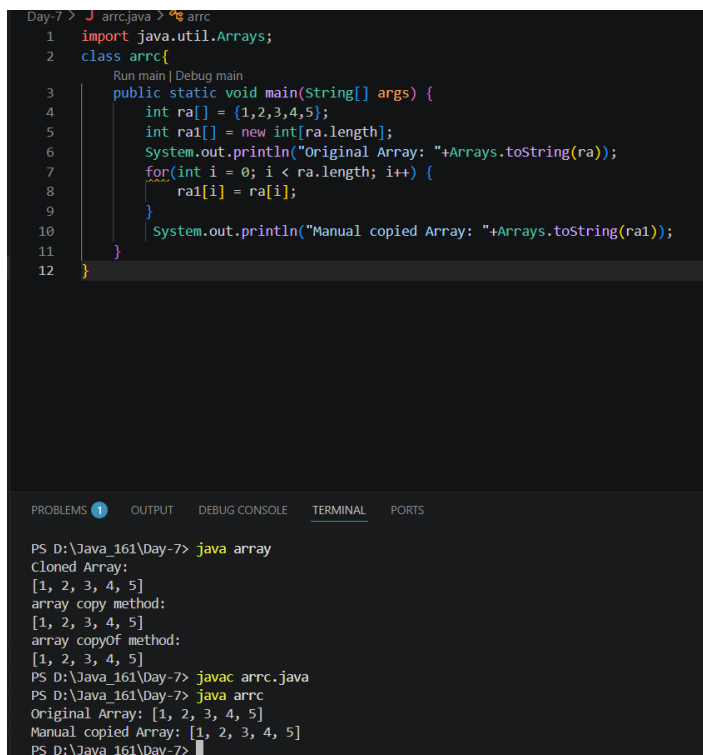
Task 5: Manual Copy

Scenario:

A shop has a list of product prices. The shop owner wants to create a backup list manually.

Write a Java program to:

- Store product prices in an array
- Copy all elements into another array using a loop
- Display the copied array



```
Day-7 > J arrc.java > %$ arrc
1  import java.util.Arrays;
2  class arrc{
    Run main | Debug main
3      public static void main(String[] args) {
4          int ra[] = {1,2,3,4,5};
5          int ral[] = new int[ra.length];
6          System.out.println("Original Array: "+Arrays.toString(ra));
7          for(int i = 0; i < ra.length; i++) {
8              ral[i] = ra[i];
9          }
10         System.out.println("Manual copied Array: "+Arrays.toString(ral));
11     }
12 }
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\Java_161\Day-7> java array
Cloned Array:
[1, 2, 3, 4, 5]
array copy method:
[1, 2, 3, 4, 5]
array copyOf method:
[1, 2, 3, 4, 5]
PS D:\Java_161\Day-7> javac arrc.java
PS D:\Java_161\Day-7> java arrc
Original Array: [1, 2, 3, 4, 5]
Manual copied Array: [1, 2, 3, 4, 5]
PS D:\Java_161\Day-7>
```